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A trilayer poly(vinylidene fluoride)/polyborate/poly(vinylidene fluoride) gel polymer electrolyte with good performance for lithium ion batteries†

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A composite membrane based on poly(vinylidene fluoride) (PVDF) and lithium polyacrylic acid oxalate borate (LiPAAOB) exhibiting high safety (self-extinguishing) and good mechanical property was prepared. The ionic conductivity of the gel polymer electrolyte (GPE) by saturating with 1 mol L^{-1} $LiPF_6$ electrolyte at ambient temperature can be up to 0.35 mS cm^{-1}, higher than that of the well-used commercial separator (Celgard 2730), 0.21 mS cm^{-1}. The lithium ion transference in the GPE at room temperature is 0.58, twice that in the commercial separator (0.27). Moreover, the GPE presents a true shut-down behavior, which is quite different from the not-real shut-down behaviour of the commercial separators. Furthermore, the absorbed electrolyte solvent is difficult to evaporate at elevated temperature. Its electrochemical performance is evaluated by using $LiFePO_4$ cathode. The obtained results suggest that this composite GPE is very attractive to large-capacity battery systems requiring high safety and low cost.

1 Introduction

Lithium ion batteries (LIBs) have gained an unprecedented importance in the last several decades as the energy storage of portable devices such as cell phones, laptop computers and so on.[1,2] Furthermore, lithium ion batteries with large capacity are also promising power sources for hybrid electric vehicles (HEVs), plug-in hybrid electric vehicles (PHEVs), pure electric vehicles (PEVs) and storage of wind, solar and tidal energy in smart grids.[3] However, large capacity battery modules require higher safety precautions due to the existence of organic liquid electrolytes which are easy to set on fire leading to occasional explosions because of misusage, short circuits or local overheating.[4] The quest for safer and more reliable electrolyte systems is therefore urgent, and solid electrolytes are promising candidates in this regard.[5–7] If solid electrolytes can replace the organic electrolytes, not only will safety be greatly improved but also other advantages such as no liquid leakage and long cycling life will be achieved.

Solid electrolytes in lithium ion batteries mainly include inorganic Li^+ ion conductors (ILICs) and polymer electrolytes (PEs). ILICs exhibit ionic conductivities in the range of 10^{-5} S cm^{-1} to 10^{-1} S cm^{-1} at room temperature and are acceptable for applications in microbatteries such as microelectronics, medical implants, the military industry, radio-frequency identification (RFID) and so on.[3,8] In the case of large capacity battery modules, their processing performance is far from satisfaction due to poor mechanical strength and sometimes high sensitivity to humidity. Ever since the work of Wright and co-workers reporting significant ion conduction in an alkali-halide poly(ethylene oxide) system,[9] PEs (solid and gel types) have attracted a great deal of attention in terms of both fundamental and applied research. In the case of applications of solid PEs, there is still some room for improvement because of low ionic conductivity mostly ranging from 10^{-8} to 10^{-5} S cm^{-1} at ambient temperature and poor mechanical strength.[10] Gel PEs (GPEs), which are formed by absorbing large amounts of liquid electrolytes in polymer matrices and have the characteristics of solid and liquid electrolytes, have gained increasing attention for their good ionic conductivity ($>10^{-4}$ S cm^{-1} at r.t.), wide electrochemical window, good thermal stability and good compatibility with both the electrodes (anode and cathode) during charge–discharge cycling.[11–13] Poly(vinylidene fluoride) (PVDF) and its copolymer poly(vinylidene fluoride-*co*-hexafluoropropylene) (P(VDF-*co*-HFP)), poly(ethylene oxide) (PEO), polyacrylonitrile (PAN) and poly(methyl methacrylate) (PMMA) have been widely studied as polymer matrices for GPEs in order to make them feasible for commercial applications.[14–25] However, the applications of GPEs in large-scale systems are currently limited because of poor mechanical strength.[26]

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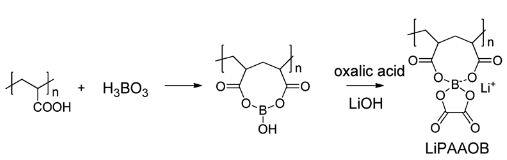
† Electronic Supplementary Information (ESI) available: The linear sweep voltammograms of Celgard 2730 and PVDF–LiPAAOB saturated with $LiFP_6$ electrolytes, the strain–stress curves for the polymers with or without the liquid electrolyte and the electrochemical performance of the electrospun PVDF membrane evaluated by using $LiFePO_4$. See DOI: 10.1039/c3ta00167a

By virtue of its attractive properties, PVDF has been chosen as a polymer host for lithium ion battery applications.[27] PVDF-based GPEs are highly electrochemically stable due to the presence of the strong electron-withdrawing functional group (–C–F) and high dielectric constant ($\varepsilon = 8.4$) which helps for larger dissolution of lithium salts and subsequent support of a high concentration of charge carriers. Many preparation methods and modifications were tried based on PVDF and its copolymers. Their conductivities are generally high enough to envisage technological applications in battery systems.[28–31] The properties of PVDF-based GPEs such as porosity, tortuosity and uniformity of pore distribution are strongly dependent on their processing methods. Compared to the GPEs prepared by methods such as solvent casting, plasticizer extraction and phase inversion, the GPEs based on electrospun membranes show higher porosity, electrolyte uptake and ionic conductivity due to the presence of a fully interconnected pore structure.[32] Electrospinning is a simple and versatile fabrication tool for preparing fibrous polymer membranes with fiber diameters ranging from several micrometers down to tens of nanometers. The GPEs based on electrospun PVDF membranes have shown their potential application in lithium ion batteries.[33,34] However, their cost is too high to accept for commercialization.

Lithium chelatoborates as electrolyte salts have been widely studied and the aim is to substitute $LiPF_6$.[35–38] The latter is unstable at elevated temperature and sensitive to trace amounts of water.[39] Among the borates, lithium bis(oxalate)borate (LiBOB) and lithium oxalydifluoroborate (LiODFB) are the promising candidates used as electrolyte salts in lithium ion batteries.[38,40] These lithium salts not only present merits such as low production cost, wide potential window and good thermal stability, but can also form effective solid electrolyte interface (SEI) films to stabilize the graphitic anode structure. Recently, we reported a single-ion conducting polymer electrolyte, lithium polyacrylic acid oxalate borate (LiPAAOB, Scheme 1), whose monomer is structurally similar to LiBOB.[41] This PE presents a single ion conductive behavior, but its ionic conductivity at ambient temperature is very low ($<10^{-5}$ S cm^{-1} at r.t.) and its mechanical strength is also poor. These drawbacks prevent its practical applications in lithium ion batteries.

In this study, we used the single-ion conductor (LiPAAOB) and electrospun PVDF membranes to form a composite membrane, PVDF–LiPAAOB, whose structure is just like trilayer separators (Celgard 2320 or Celgard 2325, polypropylene/polyethylene/polypropylene). The composite membrane having the characteristics of the electrospun PVDF (high electrolyte uptake) and the polyborate (high Li^+ ion transference number) shows a very good mechanical property and the safety is greatly increased in comparison with the commerical seperators. The ionic conductivity of the GPE based on the PVDF–LiPAAOB composite membrane at ambient temperature is higher than that for the traditional separator, Celgard 2730, saturated with $LiPF_6$ electrolyte. Moreover, the Li^+ ion transference number in the prepared GPE at room temperature is greatly increased due to the existence of the single-ion conductor. The GPE also exhibits good electrochemical performance for $LiFePO_4$. The composite membrane is very attractive for use in large-capacity lithium ion batteries.



Scheme 1 The preparation process of LiPAAOB.

2 Experimental

2.1 Synthesis of LiPAAOB

The LiPAAOB membrane was prepared according to our previous report.[41] It was synthesized by the reaction of PAA (polyacrylic acid, Aladdin, M_w: about 450 000, 200 mg), lithium hydroxide (Sinopharm, 53 mg), boric acid (Sinopharm, 78 mg) and oxalic acid (Sinopharm, 159 mg) according to Scheme 1. PAA and boric acid were dissolved in deionized water (20 mL) and heated to 80 °C for 8 h to give a homogeneous and transparent solution. Then lithium hydroxide and oxalic acid were added. The mixture was stirred at 100 °C for 24 h. After being cooled to room temperature, the solution was later cast on a flat glass plate at 50 °C to remove the solvent to obtain membranes with a thickness of about 25 μm.

2.2 Preparation of the PVDF–LiPAAOB composite membrane

Electrospun PVDF (Yingke Separators (Changzhou) Co., Ltd., 20 μm) was fixed on a glass plate and wetted with an ethanol water solution (1/1, v/v). Then LiPAAOB was put on top and wetted by the ethanol water solution (1/1, v/v). Then, another PVDF layer was put on the LiPAAOB. The composite membrane was dried at 80 °C for 5 h to give the PVDF–LiPAAOB composite membrane (thickness: 60 μm). It was punched into circular pieces ($d = 19$ mm) and then dried under vacuum at 80 °C for 48 h. After that, the pieces were soaked in an organic electrolyte (1 M $LiPF_6$ solution in EC/DMC/EMC (ethylene carbonate/dimethyl carbonate/ethyl methyl carbonate), 1/1/1, w/w/w, Zhangjiagang Guotai-Huarong New Chemical Materials Co., Ltd.) over 12 h in a glove box (water content: <1 ppm) to obtain the GPEs for further measurements.

2.3 Characterization and electrochemical evaluation

Except if stated otherwise, the following measurements were performed at room temperature. The diffuse reflectance infrared spectrometry (DRIFT) method was employed to measure infrared spectra (IR) of the samples. Before IR measurements, the samples were dried at 70 °C under vacuum for 48 h again and then stored in a glove box for 2 h. The IR spectra (4000–400 cm^{-1}, resolution 2 cm^{-1}) were recorded with a Bruker Vector-22 spectrometer. X-ray diffraction (XRD) was performed on a Bruker D8 X-ray diffractometer (50 KV, 50 mA) with a copper target ($\lambda = 1.5405$ Å) and Ni filter at a scan rate of 5° min^{-1} from 10° to 90°. The surface morphology of the prepared membranes was investigated by a Philips XL30 scanning electron microscope (SEM). The membranes were dipped into liquid nitrogen and broken into two parts, and then the SEM micrographs of the cross-sections were taken. All samples were sputtered with gold prior to SEM measurements. Thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) of the membranes were carried out by utilizing a Perkin-Elmer TGA7/DSC7. The thickness of the membranes was measured with a micrometer (SM & CTW, Shanghai). The stress–strain test was conducted using a Sansi YG832 tensile testing machine with a crosshead speed of 1 mm min^{-1}. The width of the sample was 4 mm.

The amount of liquid electrolyte uptake (η) is calculated according to eqn (1):

$$\eta = (W_t - W_0)/W_0 \times 100\% \quad (1)$$

where W_0 and W_t are the weights of the membranes before and after absorption of the organic electrolyte, respectively. The weight was weighed in a glove box.

The ionic conductivities were measured by electrochemical impedance spectroscopy (EIS). The samples were measured in blocking-type cells where the GPE membranes or the commercial separators saturated with the electrolyte were sandwiched between two stainless steel electrodes. Impedance data were obtained with an electrochemical working station CHI660C (Chenhua) in the frequency range 10 Hz–100 kHz between 25 and 75 °C. The ionic conductivity was calculated from eqn (2):

$$\sigma = l/(R_b A)\ (\mathrm{S\ cm^{-1}}) \quad (2)$$

where σ is the ionic conductivity, R_b is the bulk resistance, l is the thickness of the composite membrane or the separator, and A is the area of the stainless steel electrode since the areas of the membranes are larger than that of the steel electrode.

Linear sweep voltammograms were obtained by the electrochemical working station using a two-electrode cell. Stainless steel was used as the working electrode and lithium foil as the counter and reference electrodes. Each measurement was conducted between 0 and 6 V (*vs.* Li/Li^+) at the scan rate of 2 mV s^{-1}.

The chronoamperometry profile was obtained by the electrochemical working station measuring in blocking-type cells where the GPE membranes were sandwiched between two lithium metal electrodes. The step potential was 10 mV.

Tests of $LiFePO_4$ cathode performance using the GPE and the separator were conducted by assembling 2025 coin-type cells with lithium metal foil as the counter and reference electrodes. The $LiFePO_4$ working electrode was prepared by coating a *N*-methyl-2-pyrrolidone (NMP)-based slurry containing $LiFePO_4$ (STL Energy Technology Co., Ltd.), acetylene black and PVDF in a weight ratio of 8 : 1 : 1 on aluminium foil (thickness: 20 μm) using the doctor-blade technique, and the cast foils were then punched into circular pieces (d = 15 mm) and dried at 120 °C for 12 h under vacuum. The mass loading of $LiFePO_4$ was about 17 mg cm^{-2}. The GPE was used as the separator and electrolyte. All cells were assembled in an Ar-filled glove box. Cycling tests were carried out on a Land battery tester (CT2001A) between 2.5 and 4.2 V at 0.2 C based on the $LiFePO_4$ cathode.

3 Results and discussion

Fig. 1a shows the IR spectra of the electrospun PVDF, the prepared LiPAAOB and the PVDF–LiPAAOB composite membrane. The typical peaks of PVDF are 2980, 1400, 1075, 885, 840 and 760 cm^{-1}, attributed to $-CH_2-$ stretching, $-CH_2-$ deformation, C–C stretching, an amorphous phase band, $-CH_2-$ rocking and $-CF_2-$ bending and skeletal bending, respectively.[42,43] The peak at 1635 cm^{-1} can be assigned to the C=C absorption peak.[43] In the case of the prepared LiPAAOB, peaks at about 1735, 1650, 1445, 1420, 1340, 1190, 1050, 1020 and 655 cm^{-1} are attributed to –C=O stretching, COO– anti-symmetric stretching, COO– symmetric stretching, $-CH_n$ deformation, B–O anti-symmetric stretching, C–O–C anti-symmetric stretching, the vibration of B–O–C, C–O–C symmetric stretching and the vibration of B–O, respectively.[41] The peak at 3400 cm^{-1} is for the –OH stretching caused by absorbed water from the air due to the existence of Li^+ ions and their strong polarity. The fact that all typical peaks of PVDF and LiPAAOB appear in the spectra of the PVDF–LiPAAOB indicates the composite membrane is successfully prepared.

The X-ray diffraction (XRD) patterns of PVDF, the prepared LiPAAOB and PVDF–LiPAAOB composite are shown in Fig. 1b. It is well known that PVDF exhibits a semicrystalline structure which has a strong characteristic diffraction peak at 20.8°, corresponding to the diffraction of (110) and (200) planes of its β-phase.[44] The prepared LiPAAOB shows peaks at 14.5°, 15–27.5°, 30° and 36.7°, which are consistent with previous reports.[41] All characteristic diffraction peaks also appear in the PVDF–LiPAAOB composite membrane.

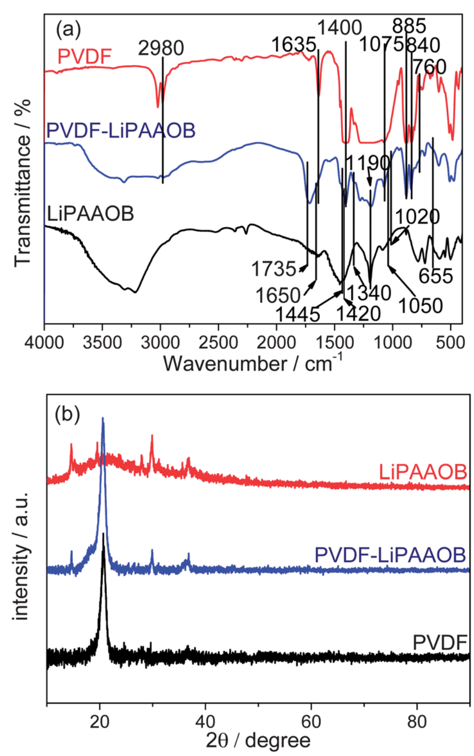


Fig. 1 (a) IR and (b) XRD spectra of the electronspun PVDF, the prepared LiPAAOB and the PVDF–LiPAAOB composite membrane.

SEM micrographs of the LiPAAOB and the PVDF–LiPAAOB composite membranes are shown in Fig. 2. From the surface of the LiPAAOB membrane (Fig. 2a), it is very rough. The inner structure of the membrane is uniform and compact from the cross-section (Fig. 2c). The compact structure imparts the membrane with a low uptake ability (about 3 wt% for propylene carbonate).[40] After forming the composite with the electrospun PVDF, porous PVDF was observed on the surface (Fig. 2b). The typical pore size is below 2 μm with a good uniformity (the inset of Fig. 2b). From the cross-section, a well formed trilayer structure can be seen, which looks likes a sandwich, and the thickness is about 60 μm. Because of the existence of the compact LiPAAOB layer between the PVDF layers, the particles of the electrode materials will not pass through the composite membrane while the Li^+ ion can transport freely, which is of great importance to prevent a possible micro short-circuit caused by the fine particles from the active electrode materials.

Thermogravimetry (TG) and differential scanning calorimetry (DSC) curves of Celgard 2730, the electrospun PVDF, the prepared LiPAAOB and the PVDF–LiPAAOB composite membrane are shown in Fig. 3a and 3b. The PVDF–LiPAAOB composite membrane is thermally stable up to 100 °C, which is about 30 °C lower than that of the commercial separator, Celgard 2730. The main reason is that the thermal stability of the composite is decided by the LiPAAOB component which slightly decomposes when the temperature reaches up to 100 °C and the main decomposition of the polymers happens at about 200 °C.[41] PVDF begins to melt at about 170 °C although no weight loss was apparent.[29] That is, the sandwiched membrane can satisfy the needs of lithium ion batteries.

The TG and DSC curves of the PVDF–LiPAAOB GPE and the Celgard 2730 separator soaked with the same amount of the electrolyte (1 mL g^{-1}) are shown in Fig. 3c and 3d. The retention ability of electrolytes is crucial to the safety of lithium ion batteries. The organic electrolyte in the commercial separator starts to evaporate at 65 °C, which is indicated by the sharp endothermic peak in the DSC curve. Its TG curve also presents a marked weight loss at the same temperature. When the

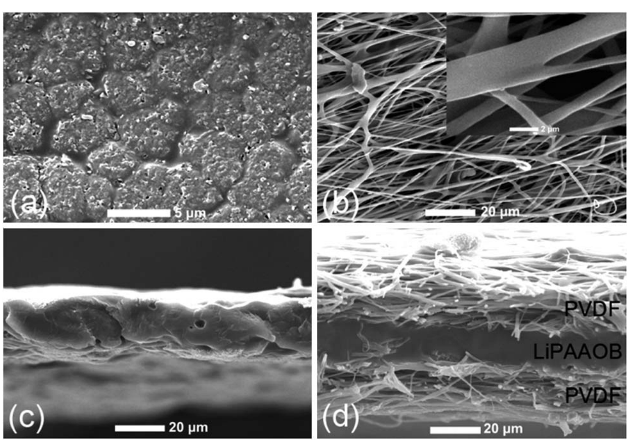


Fig. 2 SEM micrographs of (a) the surface and (c) the cross-section of the LiPAAOB membrane, (b) the surface and (d) the cross-section of the PVDF–LiPAAOB composite membrane.

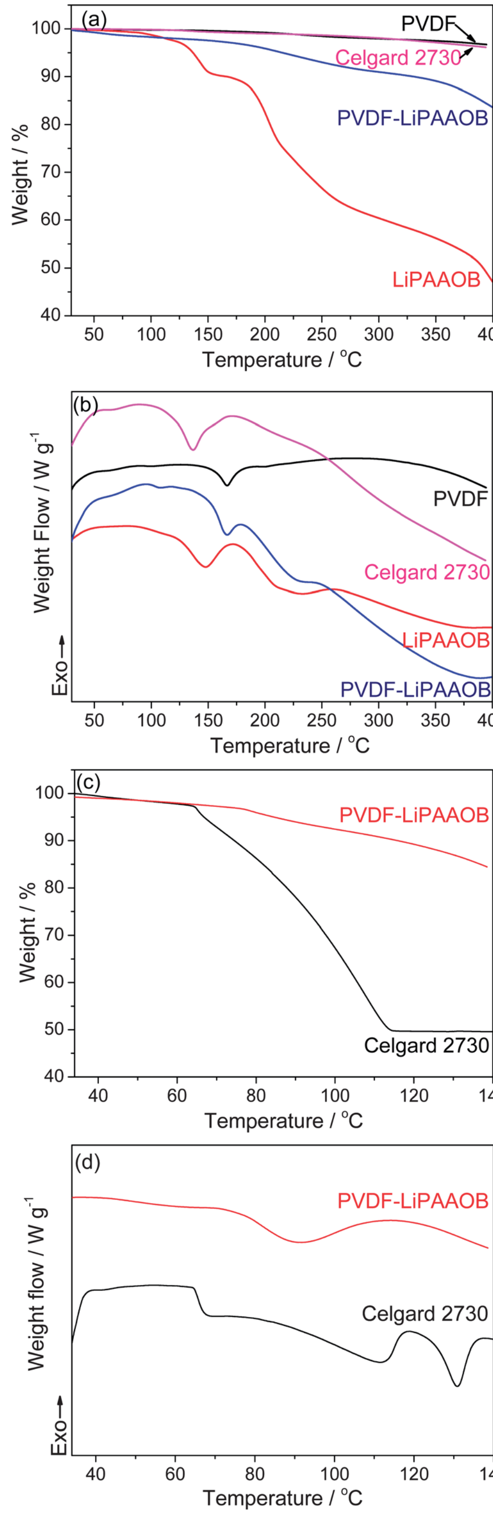


Fig. 3 (a) Thermogravimetry (TG) and (b) differential scanning calorimetry (DSC) curves of Celgard 2730, the electrospun PVDF, the prepared LiPAAOB and the PVDF–LiPAAOB composite membrane under air and the temperature rising rate is 10 °C min^{-1}, (c) TG and (d) DSC curves of Celgard 2730 and the PVDF–LiPAAOB composite after absorbing the same amount of $LiPF_6$ electrolyte (20 μL) at the temperature rising rate of 2 °C min^{-1} under nitrogen.

temperature reaches 112 °C, the evaporable solvents disappear completely. The emission of a large amount of combustible gases at temperatures below 120 °C is the main reason for a lot of accidents, such as fires and explosions, of lithium ion batteries.[29] This also suggests that at temperatures above 120 °C the lithium ion battery will not work due to very low ionic conductivity and the so-called shut-down behavior of polyethylene in the commercial separator does not help the improvement of the safety of lithium ion batteries. However, previously there has been no doubt about the true effects of this shut-down behavior since these TG and DSC behaviors with liquid electrolytes have never been tested. Furthermore, the extent of fires or explosions for lithium ion batteries evidently suggest that our finding is very important. In the case of the PVDF–LiPAAOB GPE, the absorbed electrolyte begins to evaporate at 80 °C which is 15 °C higher than that of Celgard 2730. The main reason is that the organic electrolyte is absorbed and retained by the composite membrane. Moreover, the evaporation rate is much slower than that of the commercial separator with the same amount of liquid electrolyte. When the temperature increases to 140 °C, the PVDF–LiPAAOB GPE shows only 15% weight loss which is composed of the slight decomposition of LiPAAOB and the evaporation of some of the liquid electrolyte. A large proportion of the liquid electrolyte is still retained in the gel composite membrane. The result shows that lithium ion batteries can still work at 140 °C. When the temperature is further increased, PVDF will melt down and present a different shut-down behavior, which will be of great importance to improve the practical safety of lithium ion batteries.

The combustion tests of Celgard 2730 and the PVDF–LiPAAOB composite membrane are shown in Fig. 4. The flame retarding ability of the membranes/separators is another important factor related to the safety of the lithium ion battery. When the commercial separator of Celgard 2730 was put on the flame, the separator shrank immediately and remained on fire for a short time (<3 s) (Fig. 4c). This is similar to the other traditional separators due to the existence of a combustible polyolefin matrix such as polyethylene and polypropylene. The composite membrane shows perfect flame retarding ability. It did not catch fire when put on the flame (Fig. 4d) although it also shrank. The main reason is that PVDF cannot easily set on fire and the LiPAAOB matrix is coated by the PVDF layers.

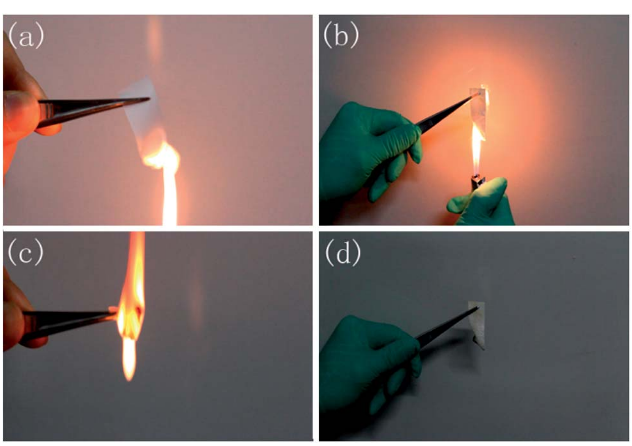


Fig. 4 The combustion test of (a and c) Celgard 2730 and (b and d) the PVDF–LiPAAOB composite membrane.

The stress–strain curves of the electrospun PVDF, the prepared LiPAAOB, the PVDF–LiPAAOB composite membrane and Celgard 2730 are shown in Fig. 5a. It is well known that poor mechanical strength limits the application of GPEs, especially in large-scale lithium ion batteries. The maxima of the stress and the strain of the PVDF–LiPAAOB composite membrane are 22.7 MPa and 14.2%, respectively, which is acceptable for lithium ion batteries. The stress value is smaller

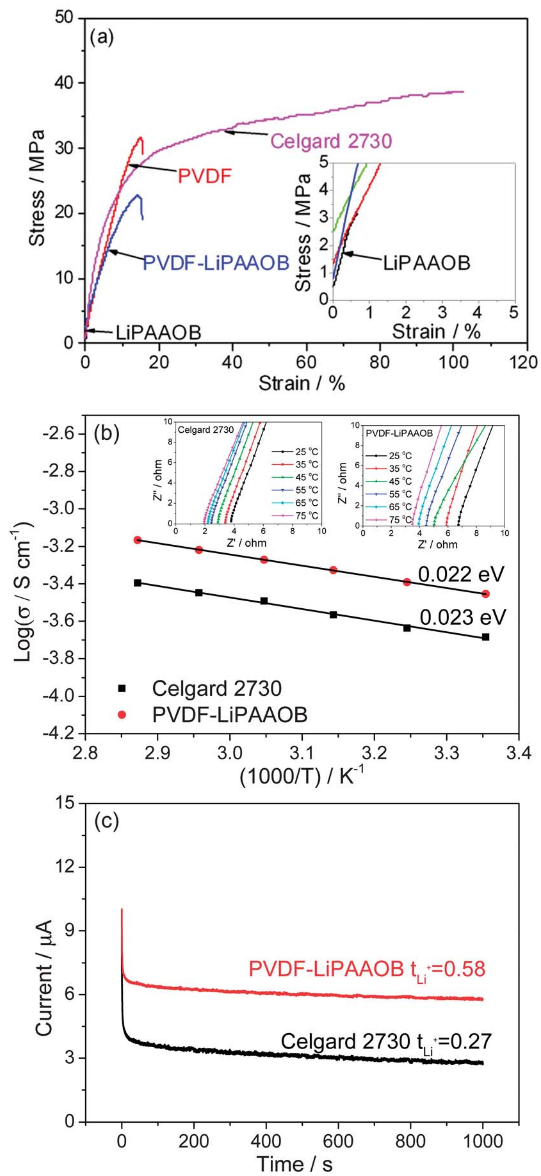


Fig. 5 (a) Stress–strain curves of Celgard 2730, the electrospun PVDF, the prepared LiPAAOB and the PVDF–LiPAAOB composite membrane; (b) impedance plots of the conductivity data at different temperatures and Arrhenius plots of the PVDF–LiPAAOB GPE and Celgard 2730 saturated with the liquid electrolyte; and (c) chronoamperometry profiles for the PVDF–LiPAAOB GPE and Celgard 2730 saturated with the liquid electrolyte at 25 °C in block cells using Li metal as both electrodes with a step potential of 10 mV.

than that of Celgard 2730. In comparison with the electrospun PVDF, the strain of the composite is decreased. The main reason is that the LiPAAOB matrix between the PVDF layers has poor mechanical strength (3 MPa in stress) and influences the strength property of the composite membrane.

The uptake amount of the composite membrane can reach 91.8 wt%, which is very close to that of Celgard 2730 (90.9 wt%). This indicates that the ionic conductivity, which is mainly dependent on the amount of the organic electrolyte, will be at least at the same order of magnitude as that of separators with liquid electrolytes.

After the uptake of the liquid organic electrolyte, it is usually acknowledged that the mechanical strength will be weaker. However, the facts do not support this supposition. The change is very little (Fig. S2†).

Fig. 5b shows the ionic conductivity dependence on the temperature for the GPE and the commercial separator (Celgard 2730) saturated with the organic liquid electrolyte over the temperature range from 25 to 75 °C. The ionic conductivity was calculated from the impedance plots shown in the insets of Fig. 5b. The typical impedance plots consist of a high frequency semicircle followed by a low frequency straight line, which correspond to the bulk/grain boundary and the electrode resistances, respectively. In the present study, the semicircular portion disappeared, indicating that the current carriers are ions and hence the total conductivity is the result of ionic conduction.[45] The resistance of the bulk electrolyte has been retrieved from the intercept of the straight line on the real axis.[46] The ionic conductivity of the PVDF–LiPAAOB GPE at 25 °C is 0.35 mS cm^{-1}, higher than that of Celgard 2730 saturated with the organic electrolyte (0.21 mS cm^{-1}). These data clearly present that the ionic conductivity of the GPEs are at the same level as the commercial separators, which is also satisfactory for practical lithium ion batteries. The dependence of ionic conductivity on temperature can be reasonably fitted by the Arrhenius eqn (3):

$$\sigma = A\exp(-E_a/RT) \tag{3}$$

where A is the pre-exponential factor and E_a is the activation energy. The calculated E_a values are 0.022 and 0.023 eV for the PVDF–LiPAAOB GPE and Celgard 2730 saturated with the liquid electrolyte, respectively. That is, the movement of ions in the gel polymer has almost the same energy as that in the organic liquid electrolytes.

The transference number of Li^+ ion estimated by chronoamperometry is shown in Fig. 5c. The major drawbacks of dual ion conducting in lithium ion batteries are the low Li^+ ion transference number and the electrodes' polarizations caused by moving anions. In addition, the mobile anions also take part in undesirable side reactions at the electrodes, which can also directly affect the performance of batteries. Therefore it is necessary to improve lithium ion transference to gain good performance for lithium ion batteries. The calculated Li^+ ion transference numbers by comparing the initial and final current values are 0.27 and 0.58 for Celgard 2730 saturated with the liquid electrolyte and the GPE, respectively. Because of the existence of a single ion conductor, the gel membrane can hinder some PF_6^- anions passing through. The LiPAAOB itself also favors the movement of Li^+ ions. In addition, the polarity of the PVDF matrix also contributes to the improvement of the Li^+ ion transference number for the GPE.

In the case of the electrochemical stability of the GPE, it is similar to that of Celgard 2730, about 4.8 V (Fig. S1 in ESI†), which is enough for lithium ion batteries.

The electrochemical performance of the PVDF–LiPAAOB GPE was evaluated by using $LiFePO_4$ as the cathode and Li metal as the counter and reference electrodes. Fig. 6 shows the comparison of the cycling behavior for the PVDF–LiPAAOB GPE and Celgard 2730 saturated with the organic electrolyte. The corresponding charge–discharge curves for selected cycles are placed as insets of the figures. The reversible capacity of $LiFePO_4$ is about 118 mA h g^{-1} at 0.2 C for the GPE (Fig. 6a), which is higher than that of the commercial separator, about 100 mA h g^{-1} (Fig. 6b). The ionic conductivity of the same magnitude and higher transference number of Li^+ ions lead the GPE to show better capacity retention than that of the commercial separator. In addition, the discharge capacity in the first cycle for Celgard 2730 saturated with the liquid electrolyte is less than the successive cycles while this does not happen in the case of the GPE. The main reason is that the $LiFePO_4$ cathode using Celgard 2730 as separator needs more time for the liquid electrolyte to disperse uniformly when both Celgard

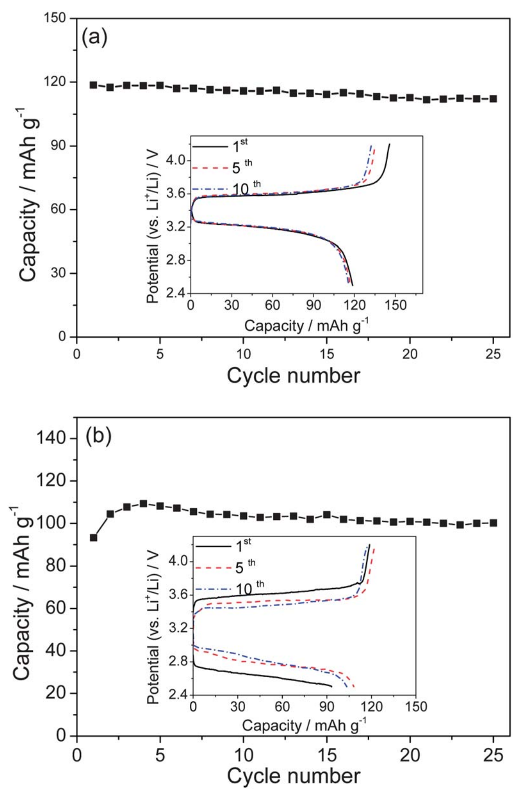


Fig. 6 Electrochemical performance of the $LiFePO_4$ cathode tested using (a) the PVDF–LiPAAOB composite membrane and (b) the Celgard 2730 separator, respectively, saturated with 1 mol L^{-1} $LiPF_6$ electrolyte and Li metal as the counter electrode.

2730 and the PVDF–LiPAAOB composite membrane are saturated with almost the same amount of the liquid electrolyte. Moreover, the $LiFePO_4$ electrode pieces used in the test are a little thicker (17 mg cm^{-2}) than most reports, such as 2 mg cm^{-2}.[47,48] This also needs a longer time to finish the formation process. After several cycles, the discharge capacity of $LiFePO_4$ using the Celgard 2730 separator becomes stable. The cycling performance for the GPE is very good, similar to that of the commercial separator. There is still no evident capacity fading during cycling. From the corresponding charge–discharge curves, typical flat-shaped voltage profiles are observed around 3.25–3.55 V in the case of the GPE (the inset of Fig. 6a), which are consistent with the reported two phase coexistence reaction for $LiFePO_4$ cathode and the difference between charge and discharge curves is very small, less than 0.4 V.[47,48] In the case of Celgard 2730, the voltage profiles of the $LiFePO_4$ are also flat (the inset of Fig. 6b). However, the difference between the charge and discharge voltages is larger, at least 0.8 V. Evidently, this higher voltage difference is due to the polarization caused by the lower transference number of Li^+ ions. In the traditional testing of the electrochemical performance of $LiFePO_4$ cathode, the cathode has less thickness and is immersed in liquid electrolytes, and the voltage difference is similar to the GPE, less than 0.4 V at low current densities.[47,48] Here both the composite membrane and the commercial separator are saturated with almost the same amount of the liquid electrolyte. The above results show that the former provides more moveable Li^+ ions, consequently the polarization and the resulting voltage difference are much smaller. Capacity retention for the GPE during cycling is very stable.

In addition, the electrochemical performance of the electrospun PVDF membrane saturated with 1 mol L^{-1} $LiPF_6$ organic electrolyte was also evaluated by using $LiFePO_4$ as the cathode and Li metal as the counter and reference electrodes (Fig. S3†). The single ion conductor, LiPAAOB, in the composite GPE increases the lithium ion transference number and the electrospun PVDF guarantees the high uptake of electrolyte and good mechanical performance of the composite GPE.

4 Conclusions

A sandwiched composite membrane consisting of a poly(vinylidene fluoride) (PVDF) and lithium polyacrylic acid oxalate borate (LiPAAOB) is prepared. The composite membrane exhibits high safety and good mechanical property. After gelation with a liquid electrolyte to achieve a gel polymer electrolyte (GPE), its ionic conductivity at ambient temperature is higher than that of the commercial separator (Celgard 2730) saturated with liquid electrolyte. Moreover, the lithium ion transference of the gel membrane at room temperature is much increased. The GPE was evaluated using $LiFePO_4$ as the cathode and Li metal as the counter and reference electrodes. The cathode delivers a higher discharge capacity at room temperature in comparison with that of the commercial separator, and good capacity retention. This gel polymer electrolyte shows great attraction to the large-scale lithium ion batteries requiring high safety and low cost.

Acknowledgements

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